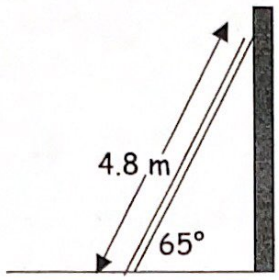
Equilibrium Worksheet#4

____
(name)

1. A uniform 4.8 m long ladder of mass 16 kg leans against a frictionless vertical wall as shown in the diagram below. a) Draw and label a free body diagram showing the forces acting on the ladder.



b) What minimum force of friction is needed at the base of the ladder to keep it from sliding?

F = __________.

2. A uniform 5.0 m ladder of mass 25 kg leans against a frictionless wall. The ladder makes an angle of 60° with the floor.

a) Calculate the force of contact against the wall

F = __________.

b) Calculate the force normal that the floor exerts on the ladder.

F = __________.

c) Calculate the friction force exerted by the floor.

F = __________.

d) Calculate the minimum coefficient of friction needed for the ladder not to slide.

μ = __________.

3. For the same ladder as above, what is the minimum angle at which the ladder will not slide out, if the coefficient of friction is 0.40?

° = __________.

Answers:

1. (b) 37N   2. (a) 71N (b) ~~245N~~ 250N (c) 71N (d) 0.29   3. 51°